Name: _____ Period: _____

IN CLASS LAB 3

Rock Cycle

How do Earth's internal and surface processes transform rocks from one type to another over time?

LEARNING INTENTION

We are learning how Earth's surface and internal processes transform rocks from one type to another so that we can use a model to explain how matter cycles through the rock cycle.

SUCCESS CRITERIA

- I can model how rocks change through weathering, erosion/deposition, compaction, heat and pressure, melting, and cooling.
- I can explain how each step of the crayon model represents a real rock cycle process.
- I can identify strengths and limitations of the model compared with the real rock cycle.

These also line up directly with the post-lab questions, especially where students match the modeled

Pre Lab:

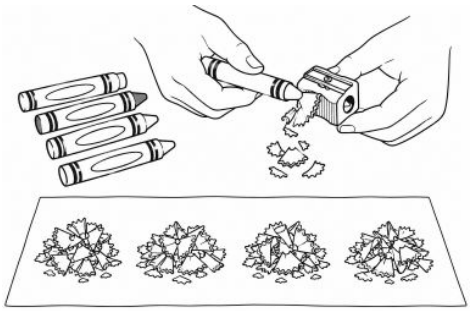
In this lab, you will model the rock cycle using crayons as your "rock material." Instead of starting with real rocks, you will use wax shavings to represent sediments, sedimentary rock, metamorphic rock, and igneous rock. Before we begin, brainstorm how you think we can use crayons to model each of the five major rock cycle processes. In your group, propose how each process could be represented.

	Explain how you may model this using crayons
Mechanical Weathering (Breaking rock into smaller pieces)	
Erosion and Deposition (Moving sediments and dropping them somewhere new)	
Compaction and Cementation / Lithification (Pressing sediments together to form rock)	
Heat and pressure / Metamorphism (Changing rock without melting it)	
Melting, Cooling, Solidification (Rock melts into magma, then cools to form igneous rock.)	

Procedure:

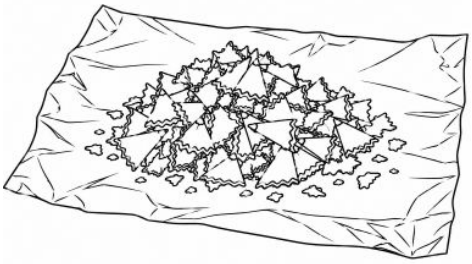
Part 1: Mechanical Weathering

1. Lay out four pieces of wax paper.
2. Each student should have two crayons of the same color.
3. Gently shave each crayon using a pencil sharpener or grater.
4. Make a small pile of shavings on each wax paper square.
5. Keep each color separate. These shavings represent rock fragments formed by weathering.



Part 2: Erosion and Deposition

1. Take the piece of aluminum foil and fold it in half.
2. Move one color of crayon shavings to the center of the foil to form a layer of sediment.
3. Add a second color on top of the first layer.
4. Continue stacking the remaining colors to create visible layers.
5. Fold the foil over the layers, leaving about ½ inch of space inside the packet. Your folded square should be about 4x4 inches





Part 3: Compaction and Cementation (Lithification)

1. Press on your fail packet to compress the layers.
2. Carefully open the foil.
3. Sedimentary rocks are weak. This one will be difficult to pick up. Try to pick up a piece to observe the layers.

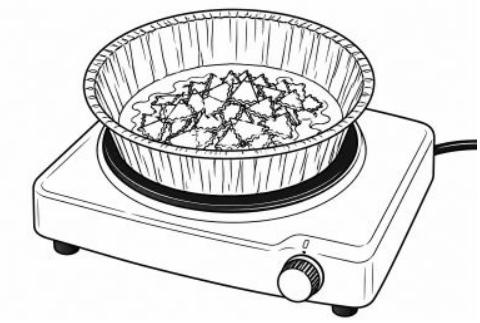
Part 4: Heat and Pressure (Metamorphism)

1. Put the pieces back into the foil and wrap it again.
2. Stand on the packet and press down firmly with your body weight. You may carefully stomp 1–3 times.
3. Open the foil and remove the “metamorphic rock.”
4. Break it in half and observe the changes in texture and layering.



Part 5: Melting, Cooling, and Solidification (Igneous Formation)

1. Place the metamorphic rock pieces into the aluminum bowl.
2. Bring the bowl to the igneous station.
3. The teacher will slowly heat the material until it melts into “magma.”
4. The melted material will be placed in an ice-water bath to cool.



Once fully cooled, remove the solid “igneous rock.”

Post Lab:

1. Match each lab step to the correct Earth process modeled.
Write the letter of the correct Earth process (A–E) on the line next to each numbered lab step. Each answer will be used only once.

_____1. Shaving crayons

_____2. Layering shavings

_____3. Pressing layers

_____4. Stomping firmly

_____5. Melting & cooling

A. Heat & pressure
B. Compaction & cementation
C. Mechanical weathering
D. Magma → solidification
E. Erosion & deposition

2. The crayon activity was used as a model of the rock cycle.

a. What real Earth processes were accurately represented by this model?

b. What important Earth processes were NOT fully represented?

c. Explain one limitation of using crayons to model the rock cycle.

3. In the real world, rock cycle processes involve interactions between Earth's spheres. Choose one rock cycle process from the lab and explain how it involves at least **two** Earth spheres (geosphere, hydrosphere, atmosphere, biosphere).

4. A crayon shaving was layered, pressed, stomped, and then melted. Describe the sequence of rock types it modeled and explain why each transformation occurred.

Cluster Set:

Base your answer(s) to the following question(s) 1-7 on the Earth Science Reference Tables, the diagram below, and your knowledge of Earth science. The photograph shows examples of older metamorphic rocks and younger volcanic rocks found in Yosemite National Park. The image on the left shows an outcrop of glacially polished quartzite interlayered with schist at May Lake. The image on the right shows Little Devils Postpile, an 8-million-year-old columnar-jointed andesite formation located near Tuolumne Meadows. Some questions may require the use of the *2024 Edition Reference Tables for Earth and Space Sciences.*

Yosemite National Park: A Landscape Shaped Over Time

Visitors to Yosemite National Park see towering rock walls, deep valleys, winding rivers, and powerful waterfalls. The cliffs rise sharply from the valley floor, and many surfaces appear smooth and curved, while other areas contain layered rock or broken fragments along the base of slopes.

Scientists have determined that the rocks exposed in Yosemite formed deep underground long before they were visible at the surface. Over time, forces within Earth pushed these rocks upward. Once exposed, rain, snow, rivers, and ice began to reshape the landscape. Water moved through cracks in the rock, sometimes freezing and expanding. Large masses of ice once covered parts of the region, carving valleys and transporting rock material to new locations.

In some lower areas, loose sediments accumulated and were later buried beneath additional material. In other places, rocks that were carried deeper underground experienced increased heat and pressure. Today, different types of rock can be found throughout the park, each recording part of Yosemite's geologic history.

The landscape of Yosemite continues to change as Earth's internal processes and surface processes interact.



_____1. Rocks in Yosemite were described as forming deep underground before becoming visible at the surface. Which type of rock most likely formed this way?

(1) Extrusive igneous rock
(2) Intrusive igneous rock
(3) Sedimentary rock
(4) Metamorphic rock

_____2. Which process most directly caused rock material in Yosemite to melt beneath Earth's surface?

(1) Energy from the Sun
(2) Surface water movement
(3) Heat from Earth's interior
(4) Plant root growth

_____3 Which statement from the passage best supports this claim?

Claim: Forces within Earth played a major role in shaping Yosemite's landscape.

(1) Visitors see waterfalls flowing through the park.
(2) Rocks were formed deep underground and later pushed upward by forces within Earth.
(3) Large cliffs rise from the valley floor.
(4) Sediments accumulated in lower areas.

4. The passage describes rocks forming deep underground and later being pushed upward over millions of years. Explain how tectonic plate movement relates to the **temporal** scale of this process.

5. The passage describes rocks forming deep underground and later becoming exposed at Earth's surface. Identify the process responsible for this vertical movement and explain how it represents a change in **spatial** scale within Earth's system.

6. Describe one interaction between the geosphere and another Earth sphere that is described in the passage.

7.Based on the passage and your knowledge of Earth science, complete the pathway below by selecting the correct row from the options below.

A rock in Yosemite originally formed as an intrusive igneous rock when _____ (A) cooled slowly beneath Earth's surface. Over time, tectonic plate movement caused _____ (B), which changed the rock's depth and brought it closer to the surface. If the rock were later buried deeper, increased heat and pressure could change it into _____ (C).

	(A)	(B)	(C)
(1)	Magma	Deposition	Sedimentary Rock
(2)	Magma	Uplift	Metamorphic Rock
(3)	Lava	Deposition	Metamorphic Rock
(4)	Lava	Uplift	Sedimentary Rock